Hydro–kinetic model for confined active rods

This document presents a non-dimensional hydro–kinetic model for a dilute suspension of self-propelled, slender rod-like swimmers confined between two rigid, stationary, parallel walls. The model consists of a kinetic (Smoluchowski) equation for the orientation distribution function, coupled self-consistently to incompressible Stokes flow through an active particle stress. We first summarize the full three-dimensional formulation, and then derive a consistent and self-contained reduction to a two-dimensional (x, z) framework.

1 Full non-dimensional hydro–kinetic formulation

1.1 Kinetic description

Let

$$\Psi = \Psi(\mathbf{x}, \mathbf{p}, t), \qquad \mathbf{x} = (x, y, z) \in \Omega \subset \mathbb{R}^3, \qquad \mathbf{p} \in S^2, \quad |\mathbf{p}| = 1,$$

denote the probability density of finding a swimmer at position $\mathbf{x}$ with orientation $\mathbf{p}$ at time t.
The non-dimensional Smoluchowski equation is written in conservative form as

$$\partial_t \Psi + \nabla \cdot \big((\mathbf{p} + \mathbf{u})\Psi - d_T \nabla \Psi\big) + \nabla_{\mathbf{p}} \cdot \big(\dot{\mathbf{p}}\, \Psi - d_R \nabla_{\mathbf{p}} \Psi\big) = 0, \tag{1}$$

where d_T and d_R are the translational and rotational diffusion coefficients, respectively.
For slender rods, the orientational drift is given by Jeffery rotation,

$$\dot{\mathbf{p}} = (\mathbf{I} - \mathbf{p}\mathbf{p}) \cdot (\nabla \mathbf{u}) \cdot \mathbf{p}. \tag{2}$$

1.2 Hydrodynamics

The surrounding fluid is governed by incompressible Stokes equations,

$$-\Delta \mathbf{u} + \nabla q = \nabla \cdot \mathbf{\Sigma}^p, \qquad \nabla \cdot \mathbf{u} = 0, \tag{3}$$

where $\mathbf{u}(\mathbf{x}, t)$ is the fluid velocity, $q(\mathbf{x}, t)$ the pressure, and $\mathbf{\Sigma}^p$ the particle-induced (active) stress.

1.3 Particle stress

The particle stress is defined directly from the orientation distribution as

$$\mathbf{\Sigma}^p(\mathbf{x}, t) = \alpha \int_{|\mathbf{p}|=1} \Psi(\mathbf{x}, \mathbf{p}, t) \left(\mathbf{p}\mathbf{p} - \frac{1}{3}\mathbf{I}\right) \mathrm{d}\mathbf{p}, \tag{4}$$

where α is the non-dimensional activity parameter. Extensile (pusher-type) swimmers correspond to $\alpha < 0$, while contractile (puller-type) swimmers correspond to $\alpha > 0$.

1.4 Boundary conditions

The suspension is confined between two rigid walls located at

$$z = \pm 1.$$

At the walls, the fluid satisfies the no-slip condition,

$$\mathbf{u} = \mathbf{0} \qquad \text{at } z = \pm 1. \tag{5}$$

The kinetic equation is supplemented by an impermeability (no-flux) condition: the normal component of the translational probability flux

$$\mathbf{J} := (\mathbf{p} + \mathbf{u})\Psi - d_T \nabla \Psi$$

vanishes at the walls,

$$\hat{\mathbf{n}} \cdot \mathbf{J} = 0 \qquad \text{at } z = \pm 1, \tag{6}$$

where $\hat{\mathbf{n}} = \pm \hat{\mathbf{z}}$ is the outward unit normal.

2 Reduction to a two-dimensional (x, z) framework

We now derive a reduced model appropriate for two-dimensional dynamics in the (x, z) plane. This reduction involves two distinct assumptions, which we state explicitly.

2.1 Spatial symmetry

First, we assume invariance in the y-direction,

$$\partial_y(\cdot) = 0,$$

and restrict the flow to be planar,

$$\mathbf{u}(x, z, t) = \begin{pmatrix} u(x, z, t) \\ w(x, z, t) \end{pmatrix}, \qquad \nabla = (\partial_x, \partial_z), \qquad \Delta = \partial_{xx} + \partial_{zz}.$$

This assumption alone does *not* restrict swimmer orientations; it only reduces the spatial dependence of the fields.

2.2 Planar-orientation restriction

To obtain a genuinely two-dimensional kinetic model, we further assume that swimmer orientations are confined to the (x, z) plane. This is a modeling choice, stronger than spatial symmetry alone, and eliminates out-of-plane orientations.
The unit director is therefore parameterized by a single angle $\theta \in [0, 2\pi)$:

$$\mathbf{p}(\theta) = \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}, \qquad \mathbf{p}_\perp(\theta) = \begin{pmatrix} -\sin\theta \\ \cos\theta \end{pmatrix}. \tag{7}$$

With this assumption, the orientation distribution reduces to

$$\Psi = \Psi(x, z, \theta, t),$$

with orientations living on the circle S^1.

3 Reduced 2D hydro–kinetic model

3.1 Kinetic equation

Under the above assumptions, the Smoluchowski equation becomes

$$\partial_t \Psi + \nabla \cdot \big((\mathbf{p} + \mathbf{u})\Psi - d_T \nabla \Psi\big) + \partial_\theta \big(\dot{\theta}\, \Psi - d_R \partial_\theta \Psi\big) = 0, \tag{8}$$

where the angular drift is

$$\dot{\theta} = \mathbf{p}_\perp(\theta) \cdot (\nabla \mathbf{u})\, \mathbf{p}(\theta), \qquad \nabla \mathbf{u} = \begin{pmatrix} \partial_x u & \partial_z u \\ \partial_x w & \partial_z w \end{pmatrix}. \tag{9}$$

3.2 Hydrodynamics

The incompressible Stokes equations retain their form in the reduced geometry,

$$-\Delta \mathbf{u} + \nabla q = \nabla \cdot \mathbf{\Sigma}^p, \qquad \nabla \cdot \mathbf{u} = 0. \tag{10}$$

3.3 Purely 2D traceless particle stress

To maintain consistency with the planar reduction, we define a purely two-dimensional, traceless particle stress directly from the orientation distribution:

$$\mathbf{\Sigma}^p(x, z, t) = \alpha \int_0^{2\pi} \Psi(x, z, \theta, t) \left(\mathbf{p}(\theta)\mathbf{p}(\theta) - \frac{1}{2}\mathbf{I}_2 \right) \mathrm{d}\theta, \tag{11}$$

where $\mathbf{I}_2$ is the 2×2 identity tensor. This definition preserves tracelessness in two dimensions and avoids the introduction of coarse-grained moment fields.

3.4 Boundary conditions

At the rigid walls $z = \pm 1$, the no-slip condition reads

$$u = w = 0 \qquad \text{at } z = \pm 1. \tag{12}$$

The no-flux condition reduces to a Robin boundary condition for Ψ. Since $\hat{\mathbf{z}} \cdot \mathbf{p}(\theta) = \sin\theta$ and $w = 0$ at the wall,

$$d_T\, \partial_z \Psi(x, \pm 1, \theta, t) = \sin\theta\, \Psi(x, \pm 1, \theta, t). \tag{13}$$

4 Component-wise summary in the (x, z) framework

For clarity, we summarize the reduced model in component form, respecting the reduced orientation distribution:

$$\Psi = \Psi(x, z, \theta, t),$$

Kinetic equation:

$$\begin{aligned}\partial_t \Psi + \partial_x[(\cos\theta + u)\Psi - d_T\,\partial_x\Psi] + \partial_z[(\sin\theta + w)\Psi - d_T\,\partial_z\Psi] \\ + \partial_\theta\Big[\dot{\theta}\,\Psi - d_R\,\partial_\theta\Psi\Big] = 0,\end{aligned} \tag{14}$$

with

$$\dot{\theta} = -\sin\theta\,(\partial_x u\cos\theta + \partial_z u\sin\theta) + \cos\theta\,(\partial_x w\cos\theta + \partial_z w\sin\theta).$$

Stokes equations:

$$-\Delta u + \partial_x q = \partial_x \Sigma^p_{xx} + \partial_z \Sigma^p_{xz}, \tag{15}$$

$$-\Delta w + \partial_z q = \partial_x \Sigma^p_{zx} + \partial_z \Sigma^p_{zz}, \tag{16}$$

$$\partial_x u + \partial_z w = 0. \tag{17}$$

2D traceless particle stress components. With the purely two-dimensional traceless definition, the individual stress components are

$$\Sigma^p_{xx} = \alpha \int_0^{2\pi} \Psi\left(\cos^2\theta - \tfrac{1}{2}\right)\,d\theta = \frac{\alpha}{2}\int_0^{2\pi} \Psi\,\cos(2\theta)\,d\theta, \tag{18}$$

$$\Sigma^p_{zz} = \alpha \int_0^{2\pi} \Psi\left(\sin^2\theta - \tfrac{1}{2}\right)\,d\theta = -\frac{\alpha}{2}\int_0^{2\pi} \Psi\,\cos(2\theta)\,d\theta, \tag{19}$$

$$\Sigma^p_{xz} = \Sigma^p_{zx} = \alpha \int_0^{2\pi} \Psi\,\cos\theta\,\sin\theta\,d\theta = \frac{\alpha}{2}\int_0^{2\pi} \Psi\,\sin(2\theta)\,d\theta. \tag{20}$$

By construction, the stress is traceless,

$$\Sigma^p_{xx} + \Sigma^p_{zz} = 0,$$

so that the isotropic contribution is absorbed into the pressure.

Boundary conditions at $z = \pm 1$:

$$u = w = 0, \qquad d_T\,\partial_z\Psi = \sin\theta\,\Psi.$$

Equations (8)–(13) define a closed, non-dimensional, two-dimensional hydro–kinetic model for confined active rods with planar orientations. This formulation provides a clean and consistent starting point for analytical reduction, linear stability analysis, or numerical implementation.

5 Steady base states for the first and second transitions

Simulations indicate that the confined active suspension can admit distinct equilibrium states depending on the activity α and diffusion parameters (d_T, d_R). The stability analysis developed later quantifies how small perturbations evolve away from a chosen equilibrium (base state). The present focus is on two equilibria: (i) a quiescent state (no mean flow), and (ii) a unidirectional-flow state. The loss of stability of the quiescent base is referred to as the *first transition*, leading to a unidirectional-flow base state. The loss of stability of the unidirectional base to bending-type instabilities is referred to as the *second transition*.

5.1 1D unidirectional base-state ansatz

Starting from the 2D hydro–kinetic model (Eqs. (8)–(13), or equivalently (14)–(20)), we impose a one-dimensional, steady, unidirectional-flow reduction:

$$\partial_t(\cdot) = 0, \qquad \partial_x(\cdot) = 0, \qquad \mathbf{u}(z) = (U(z),\, 0), \qquad S(z) := \partial_z U(z), \tag{21}$$

with no-slip at the walls $U(\pm 1) = 0$, and the orientation distribution

$$\Psi = \Psi(z, \theta), \qquad \theta \in [0, 2\pi) \text{ periodic.}$$

This framework contains the quiescent equilibrium as the subcase

$$U(z) \equiv 0 \quad \Longleftrightarrow \quad S(z) \equiv 0. \tag{22}$$

5.2 Jeffery drift under 1D unidirectional shear

For $\mathbf{u} = (U(z), 0)$, the velocity gradient has the only nonzero component $\partial_z u = S(z)$. Substituting into the 2D angular drift formula $\dot{\theta} = \mathbf{p}_\perp \cdot (\nabla \mathbf{u})\mathbf{p}$ yields

$$\dot{\theta} = -\, S(z)\, \sin^2\theta. \tag{23}$$

5.3 Reduced steady kinetic equation (closed form)

Using (21)–(23) in the 2D Smoluchowski equation (8) (or component form (14)), the steady kinetic equation reduces to

$$\sin\theta\, \partial_z \Psi(\theta, z) - \; S(z)\, \partial_\theta\big(\sin^2\theta\, \Psi(\theta, z)\big) = \; d_T\, \partial_{zz}\Psi(\theta, z) + \; d_R\, \partial_{\theta\theta}\Psi(\theta, z), \tag{24}$$

with no-flux boundary conditions at $z = \pm 1$:

$$\big(\sin\theta\, \Psi - d_T\, \partial_z \Psi\big)\Big|_{z=\pm 1} = 0, \qquad \Psi(\theta + 2\pi, z) = \Psi(\theta, z). \tag{25}$$

The quiescent base state corresponds to $S \equiv 0$, in which case (24) reduces to a linear elliptic problem in (z, θ).

5.4 Active shear stress and shear–stress closure

Under the same 1D reduction, the only particle-stress component required to determine the unidirectional shear is the active shear stress

$$\Sigma_{xz}^{(p)}(z) = \alpha \int_0^{2\pi} \cos\theta \, \sin\theta \, \Psi(\theta, z) \, d\theta. \tag{26}$$

The streamwise Stokes balance (Eq. (15) with $\partial_x(\cdot) = 0$) reduces to

$$-\partial_{zz} U(z) = \partial_z \Sigma_{xz}^{(p)}(z).$$

Integrating once in z gives a relation between the shear rate $S(z) = U'(z)$ and the shear stress:

$$S(z) = -\Sigma_{xz}^{(p)}(z) + C,$$

where C is an integration constant. Using no-slip, $U(\pm 1) = 0$, implies the zero-mean constraint

$$\int_{-1}^{1} S(z) \, dz = U(1) - U(-1) = 0,$$

which fixes

$$C = \frac{1}{2} \int_{-1}^{1} \Sigma_{xz}^{(p)}(\xi) \, d\xi.$$

Therefore the shear–stress closure may be written as

$$S(z) = -\Sigma_{xz}^{(p)}(z) + \frac{1}{2} \int_{-1}^{1} \Sigma_{xz}^{(p)}(\xi) \, d\xi, \qquad \int_{-1}^{1} S(z) \, dz = 0. \tag{27}$$

5.5 Connection to the Galerkin steady solver

Equations (24)–(27) form a closed nonlinear steady system for the unknowns $\Psi(\theta, z)$ and $S(z)$ (equivalently $U(z)$), and coincide with the steady 1D kinetic model and shear–stress closure used in the quadrature-based Galerkin formulation employed to compute steady equilibria. Once a steady solution is computed, it provides the base state Ψ_0 (and U_0, S_0) for the subsequent linear stability analysis of either the first transition (from quiescent) or the second transition (from unidirectional flow).

6 Linearized model about a general steady base state

We now derive a linearized hydro–kinetic model describing the evolution of small perturbations about a *general steady equilibrium* of the confined active fluid. The equilibrium is taken to be a one-dimensional, steady solution of the reduced unidirectional-flow problem derived in Section 5 and computed using the quadrature-based Galerkin formulation.

6.1 General steady base state

Let

$$\mathbf{u}_0(z) = (U_0(z),\, 0), \qquad S_0(z) := \partial_z U_0(z), \qquad \Psi_0 = \Psi_0(z,\theta), \tag{28}$$

denote a steady solution of the one-dimensional system (24)–(27), satisfying the wall boundary conditions and normalization constraints. This formulation encompasses both equilibrium states of interest:

- *Quiescent equilibrium (first transition):*

$$U_0(z) \equiv 0, \qquad S_0(z) \equiv 0.$$

- *Unidirectional-flow equilibrium (second transition):*

$$U_0(z) \neq 0, \qquad S_0(z) = \partial_z U_0(z) \neq 0.$$

6.2 Perturbation ansatz

Introduce a small parameter $\varepsilon \ll 1$ and decompose all fields as

$$\begin{aligned}
\Psi(x,z,\theta,t) &= \Psi_0(z,\theta) + \varepsilon\, \Psi'(x,z,\theta,t) + O(\varepsilon^2),\\
u(x,z,t) &= U_0(z) + \varepsilon\, u'(x,z,t) + O(\varepsilon^2),\\
w(x,z,t) &= 0 + \varepsilon\, w'(x,z,t) + O(\varepsilon^2),\\
q(x,z,t) &= q_0(z) + \varepsilon\, q'(x,z,t) + O(\varepsilon^2),
\end{aligned} \tag{29}$$

where $q_0(z)$ absorbs the base-state normal stress and does not influence the perturbation dynamics.

6.3 Linearized kinetic equation

Substituting (29) into the full kinetic equation (8) and retaining $O(\varepsilon)$ terms yields

$$\begin{aligned}
\partial_t \Psi' &+ \partial_x\big[(\cos\theta + U_0)\Psi' + u'\Psi_0 - d_T\, \partial_x \Psi'\big]\\
&+ \partial_z\big[\sin\theta\, \Psi' + w'\Psi_0 - d_T\, \partial_z \Psi'\big]\\
&+ \partial_\theta\Big[\dot\theta_0\, \Psi' + \dot\theta'\, \Psi_0 - d_R\, \partial_\theta \Psi'\Big] = 0,
\end{aligned} \tag{30}$$

where the base-state angular drift is

$$\dot\theta_0(z,\theta) = -\, S_0(z)\, \sin^2\theta, \tag{31}$$

and the perturbation-induced angular drift is

$$\dot\theta' = -\sin\theta\big(\partial_x u' \cos\theta + \partial_z u' \sin\theta\big) + \cos\theta\big(\partial_x w' \cos\theta + \partial_z w' \sin\theta\big)\,. \tag{32}$$

6.4 Linearized Stokes equations

The perturbation velocity satisfies the forced incompressible Stokes equations

$$-\Delta u' + \partial_x q' = \partial_x \Sigma^{p\,\prime}_{xx} + \partial_z \Sigma^{p\,\prime}_{xz}, \tag{33}$$

$$-\Delta w' + \partial_z q' = \partial_x \Sigma^{p\,\prime}_{zx} + \partial_z \Sigma^{p\,\prime}_{zz}, \tag{34}$$

$$\partial_x u' + \partial_z w' = 0, \tag{35}$$

with $\Delta = \partial_{xx} + \partial_{zz}$.

6.5 Linearized particle stress

Since the particle stress depends linearly on the distribution function, its perturbation depends only on Ψ':

$$\Sigma^{p\,\prime}(x,z,t) = \alpha \int_0^{2\pi} \Psi'(x,z,\theta,t) \left(\mathbf{p}(\theta)\mathbf{p}(\theta) - \tfrac{1}{2} I_2 \right) d\theta, \qquad \mathbf{p}(\theta) = (\cos\theta, \sin\theta). \tag{36}$$

Equivalently, the component-wise expressions are

$$\Sigma^{p\,\prime}_{xx} = \frac{\alpha}{2} \int_0^{2\pi} \Psi'(x,z,\theta,t) \cos(2\theta)\, d\theta, \tag{37}$$

$$\Sigma^{p\,\prime}_{zz} = -\frac{\alpha}{2} \int_0^{2\pi} \Psi'(x,z,\theta,t) \cos(2\theta)\, d\theta, \tag{38}$$

$$\Sigma^{p\,\prime}_{xz} = \Sigma^{p\,\prime}_{zx} = \frac{\alpha}{2} \int_0^{2\pi} \Psi'(x,z,\theta,t) \sin(2\theta)\, d\theta, \tag{39}$$

with $\Sigma^{p\,\prime}_{xx} + \Sigma^{p\,\prime}_{zz} = 0$.

6.6 Linearized boundary conditions

At the rigid walls $z = \pm 1$, the no-slip condition gives

$$u'(x, \pm 1, t) = 0, \qquad w'(x, \pm 1, t) = 0, \tag{40}$$

and the impermeability condition linearizes to

$$d_T\, \partial_z \Psi'(x, \pm 1, \theta, t) = \sin\theta\, \Psi'(x, \pm 1, \theta, t). \tag{41}$$

6.7 Role in the stability analysis

The system (30)–(41) defines a generalized linear stability problem about an arbitrary steady base state (Ψ_0, U_0) obtained from the 1D Galerkin solver. Specializing this formulation to the quiescent equilibrium characterizes the first transition, while specializing to a unidirectional-flow equilibrium characterizes the second transition.

6.8 Normal-mode reduction and eigenvalue formulation

The linearized perturbation fields are sought in normal-mode form, with a single streamwise Fourier wavenumber k and exponential time dependence:

$$(\Psi', u', w', q', \Sigma^{(p)\prime})(x, z, \theta, t) = (\hat{\Psi}, \hat{u}, \hat{w}, \hat{q}, \widehat{\Sigma}^{(p)})(z, \theta)\, \exp(ikx + \gamma t), \tag{42}$$

where $\gamma \in \mathbb{C}$ is the growth rate (eigenvalue), and $\hat{\Psi}(z, \theta)$ is 2π-periodic in θ.
Under (42), derivatives in x and t reduce to

$$\partial_t \mapsto \gamma, \qquad \partial_x \mapsto ik, \qquad \partial_{xx} \mapsto -k^2, \tag{43}$$

so that the Laplacian in the Stokes operator becomes

$$\Delta = \partial_{xx} + \partial_{zz} \;\mapsto\; \partial_{zz} - k^2. \tag{44}$$

Hence, the linearized system (Eqs. 30–41) reduces to a generalized eigenvalue problem in the remaining independent variables (z, θ):

$$\mathcal{L}(k; \Psi_0, U_0, S_0)\, \hat{\mathbf{q}} = \gamma\, \mathcal{M}\, \hat{\mathbf{q}}, \tag{45}$$

where $\hat{\mathbf{q}}$ denotes the collection of unknown modal amplitudes, $\mathcal{L}$ is a linear differential operator in (z, θ) whose coefficients depend on the steady base state $\Psi_0(z, \theta)$, $U_0(z)$, and $S_0(z) = \partial_z U_0(z)$, and $\mathcal{M}$ is the eigenvalue mass operator (nonzero only in the kinetic equation for $\hat{\Psi}$).
The boundary conditions inherit the same normal-mode form. In particular, at the walls $z = \pm 1$,

$$\hat{u} = \hat{w} = 0, \qquad d_T\, \partial_z \hat{\Psi} = \sin\theta\, \hat{\Psi} \qquad \text{at } z = \pm 1. \tag{46}$$

Because θ remains periodic, no boundary conditions are imposed in θ; Fourier representations will be used in that variable.

Implementation note (Dedalus). In Dedalus, one may therefore solve the EVP (45) by working directly on the (z, θ) domain, treating k as a parameter and using Fourier expansions in θ and Chebyshev expansions in z. This avoids including x as an explicit coordinate and isolates a single wavenumber k without post-processing of multiple Fourier blocks.